Projection Modes
It is desirable that the projector possess varied projection modes to accommodate ceiling mounting and rear projection orientations. Usually, a ceiling mounted projector needs to have vertical inversion of the image to counter the upside-down nature of ceiling mounting. The rear projection modes require the projected image to be inverted horizontally.

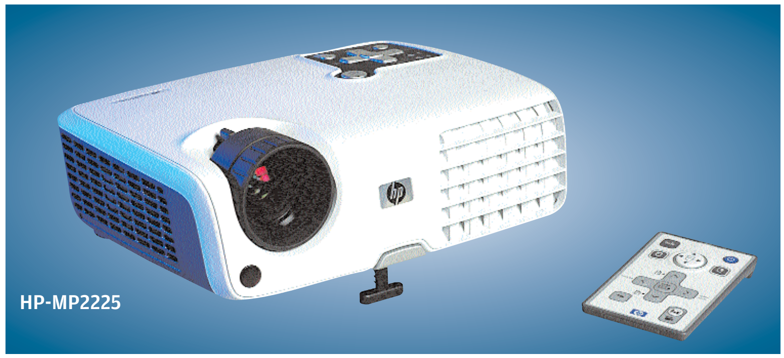
Keystone Correction
This feature is used to correct the trapezoidal distortion in the geometry of the projected image. It happens when the projector isn't directly in front of the screen. Ideally, any form of keystone correction should be avoided, because it stretches and skews an image for correction, thus bringing about degradation of image quality.

Number Of Preset Display Modes
These come in handy when you need speed and convenience over quality. Presets allow you to flick between the high-brightness Presentation mode and the totally different setting required for movies.

The Remote Control
A remote control with a full range of controls is not just a feature but a necessity when it comes to a projector, considering the nature of its placement. A remote-equipped laser pointer and mouse is a welcome bonus, which can make presentations a lot easier.

The Contenders
Of the eight projectors, three were XGA (1024 x 768) and the rest were SVGA (800 x 600). We classified them into two categories based on their native resolutions. The present lot reflect the improvement in LCD and DLP projection technologies, which have blurred the lines between their stereotypical strengths and weaknesses. But still, they weren't without their share of faults that low-cost projectors are notorious for.

Features
SVGA
Their low price notwithstanding, these were surprisingly well-built, with an impressive list of features earlier available only on the pricier models. Today, you can buy a projector with a 2000 lumens brightness rating in this entry-level SVGA bracket.



HP-MP2225

How We Tested

The Test Bed
The test bed had an Intel Pentium 4 3.2 GHz processor, an MSI 875P NEO motherboard, 1 GB DDR 400 SDRAM and an ATi X850-based video card. Windows XP Professional with Service Pack 2 and 32-bit colour depth was used, with the Windows and ATi-based graphics enhancements and presets turned off.

Evaluation Pre-Setup
The tests were carried out in absolute darkness with zero ambient light. In order to keep the evaluation process fair, we maintained a uniform size of the projected image. Having a fixed distance would have been unfair, since projectors have different throw distances in accordance to the room size specifications they are built for. Zoom was set to zero, and great pains were taken to do away with keystone correction altogether, unless unavoidable, because doing that reduces image quality. DVI interconnects were given precedence over D-SUB wherever available. We gave the projector lamp 15 minutes more than its rated warm-up time to ensure pure colour rendition, after which the pre-test calibration was executed. All projectors were tested at their native resolutions with a default refresh rate of 75 Hz. Attempts were made to rectify tracking or phase errors, if any. We avoided using the default projector settings and instead calibrated each projector to give the best fidelity possible. A white projection screen 90x72 inches large was used for the tests.

Features
Here we focused on key features such as brightness, contrast ratio, optical zoom ratio, keystone, preset modes, fine picture and colour temperature control etc. Interconnect availability such as DVI, D-SUB, and S-Video, was also considered.

Another key feature is the weight and dimensions of the projector, as business applications need to consider portability due to need for constant mobility. Special features such as USB control port, an inbuilt mouse, and remotes with Laser pointers were given additional points.

The Tests
DisplayMate Video Edition
DisplayMate is a leading video evaluation tool, which employs accurate proprietary test patterns to evaluate image quality, as also to reveal defects in video displays. We tested the projectors using 26 relevant parameters such as contrast, brightness, focussing, resolution, moiré, colour gradation accuracy, purity, streaking, black and white levels, blooming, ghosting, flicker, etc. This gives a definitive idea of the resolution accuracy and colour rendition, and also uncovers serious flaws which could easily be missed otherwise.

Picture Quality
Here we used our reference high resolution bitmaps and Adobe Photoshop PSD files to gauge the colour rendition and the detail levels. The test images consisted of a variety of real photographs and flat-shaded ones with less colour complexity. Each image had a characteristic feature which formed the basis for benchmarking. These were instrumental in gauging grille effect, colour gradation discrepancies, impurity and most importantly, the black level detail of the projector.

Presentation Test
This test included a PowerPoint presentation with pictures, graphs, and text of varying font sizes and colours. Our main focus was the readability of normal and inverted text, with the graph and photo quality coming second.

Animation Test
For the animation test, we used professional reference grade, imported DVDs of *Shrek* and *Spirit*. *Shrek* was used to test the intensity, colour-scaling, and the ability of the projector